

CSCE 2004 –Programming Foundations I
Fall 2025
University of Arkansas, Fayetteville
Assignment 2 –Due 09/12/2025

1. Problem Statement:

The goal of this programming assignment is to give students experience using conditional statements in C++. You will be designing a basic “Academic Advising System”. The student will input their major and ask if they had taken few key courses. Based on this information, the system will suggest which course(s) they should take next.

2. Design:

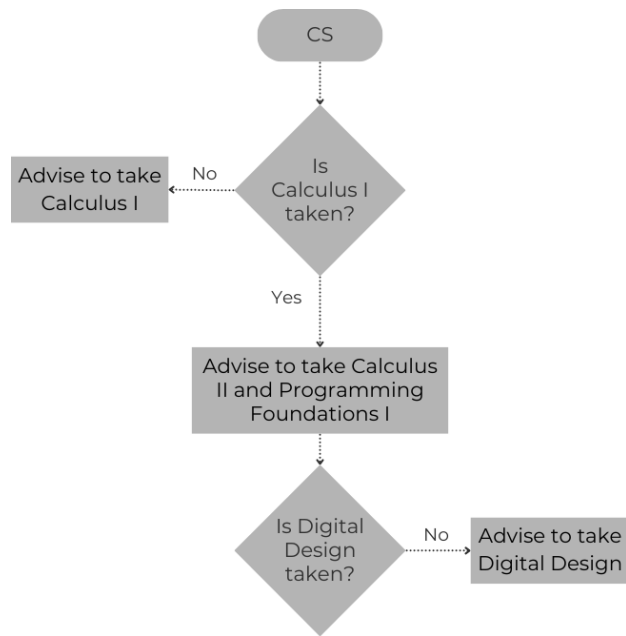
Input/Output Handling

- Display a menu for the user to choose their major by entering a number:
 - 1 → Electrical Engineering (EE)
 - 2 → Computer Science (CS)
 - 3 → Computer Engineering (CE)
- Prompt the user to enter if they have taken the following courses (Y/N):
 - Calculus I
 - Digital Design

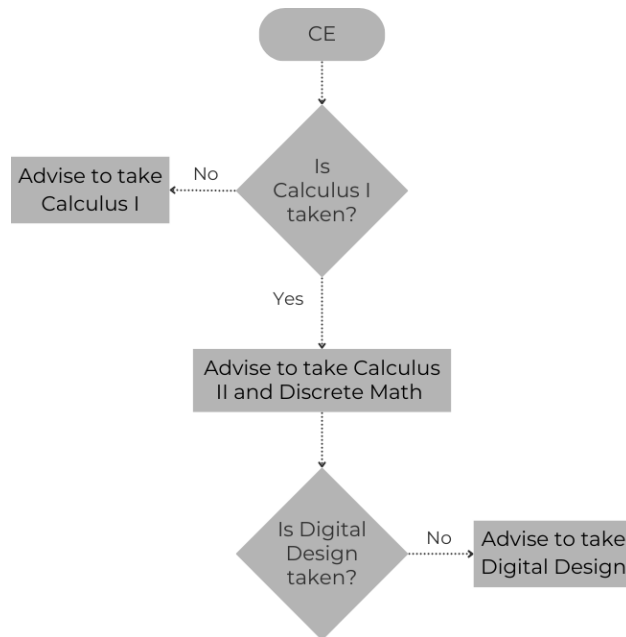
Requirements

- No loops or functions should be used. The interaction is single-use and straightforward.
- Use a **switch statement** to handle logic for different majors.
- Use **if-else statements** inside each case to suggest the next course(s) they should take based on the given information.

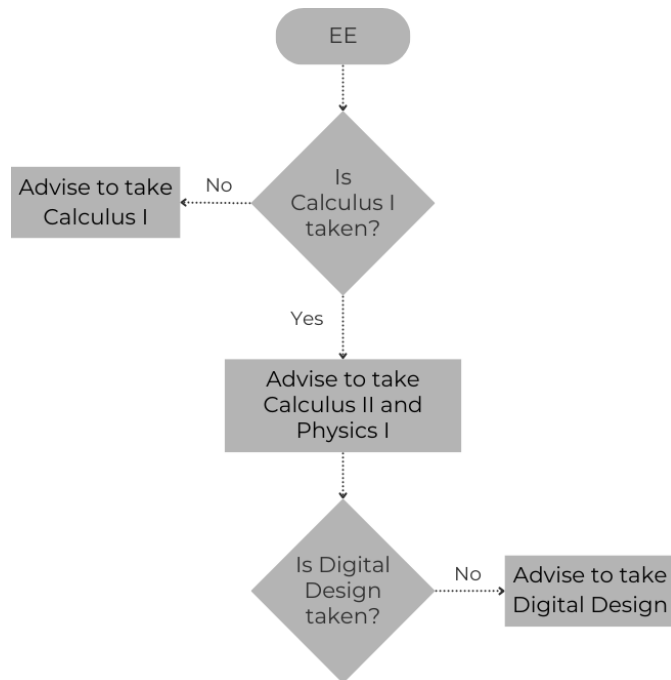
CS majors:



CE majors:



EE majors:



3. Implementation:

Since you are starting with a “blank piece of paper” to implement this project, it is very important to develop your code incrementally writing comments, adding code, compiling, debugging, a little bit at a time. This way, you always have a program that “does something” even if it is not complete.

As a first step, it is always a good idea to start with an empty main function, and add the code to read input data from the user, and implement one of the cases. Once this part is working, you can do the rest of the code.

4. Testing:

Test your program with a variety of inputs to check that it operates correctly for all of the requirements listed above.

Include screenshots of testing all the three menu functions and the invalid cases in your project report.

5. Documentation:

When you have completed your C++ program, write a short report using the “Programming Project Report Template” describing what the objectives were, what you did, and the status of

the program. Does your program work properly for all test cases? Are there any known problems? Save this project report in a separate document to be submitted electronically.

6. Sample Outputs:

Sample 1 -

```
Select your major
1. Electrical Engineering (EE)
2. Computer Science (CS)
3. Computer Engineering (CE)
-----
Enter your choice (1-3): 1
Have you taken Calculus I (Y/N): y
Have you taken Digital Design (Y/N): n

--- Advising ---
You should consider taking Calculus II and Physics.
You should take Digital Design.
-----
```

Sample 2 -

```
Select your major
1. Electrical Engineering (EE)
2. Computer Science (CS)
3. Computer Engineering (CE)
-----
Enter your choice (1-3): 2
Have you taken Calculus I (Y/N): n
Have you taken Digital Design (Y/N): y

--- Advising ---
You should take Calculus I.
```

Sample 3-

```
Select your major
  1. Electrical Engineering (EE)
  2. Computer Science (CS)
  3. Computer Engineering (CE)
-----
Enter your choice (1-3): 3
Have you taken Calculus I (Y/N): Y
Have you taken Digital Design (Y/N): Y

--- Advising ---
You should consider taking Calculus II and Programming Foundations I.
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```

7. Project Submission:

In this class, we will be using electronic project submission to make sure that all students hand their programming projects and labs on time, and to perform automatic plagiarism analysis of all programs that are submitted.

When you have completed the tasks above, go to Blackboard to upload your documentation (a single docx or pdf file), and your C++ program (a single cpp file). Do NOT upload an executable version of your program.

The dates on your electronic submission will be used to verify that you met the due date above. All late projects will receive reduced credit:

10% off if less than 1 day late,
20% off if less than 2 days late, 30% off if
less than 3 days late, no credit if more than
3 days late.

You will receive partial credit for all programs that compile even if they do not meet all program requirements, so handing projects in on time is highly recommended.

8. Academic Honesty Statement:

Students are expected to submit their own work on all programming projects, unless group projects have been explicitly assigned. Students are NOT allowed to distribute code to each other, or copy code from another individual or website. Students ARE allowed to use any materials on the class website, or in the textbook, or ask the instructor and/or GTAs for assistance.

This course will be using highly effective program comparison software to calculate the similarity of all programs to each other, and to homework assignments from previous semesters. Please do not be tempted to plagiarize from another student.

Violations of the policies above will be reported to the Provost's office and may result in a ZERO on the programming project, an F in the class, or suspension from the university, depending on the severity of the violation and any history of prior violations.